

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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Finding the right CWE page

This report adds search over the CWE book. It continues report 3.

What we did

Given a Java file, we turn the risky part into a short question and search the book.

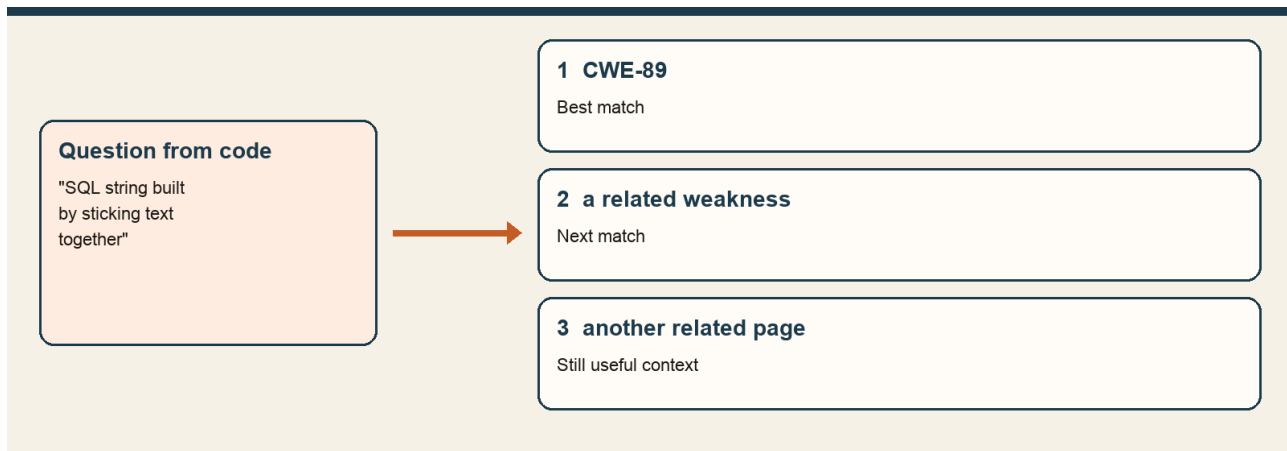
We try word match, a small meaning model called MiniLM, and a mix of both. The mix also uses the simple checker's hint and related CWE pages.

On 18 practice questions, MiniLM put the right CWE first about 94 times out of 100 ($R@1 = 0.944$). Word match did this about 78 times out of 100.

$R@1$ means the right page was the first result. MiniLM was first on about 17 of the 18 questions. 17 divided by 18 is 0.944.

On this small set the mix matched MiniLM. It did not beat it yet. If the meaning model is missing, we fall back to word match and we record that.

Picture



Numbers for this step only

Search	Right page first ($R@1$)	Questions
Word match (TF-IDF)	0.778	18
Meaning match (MiniLM)	0.944	18
Mix of both	0.944	18

What is next

Next we will fix the shape of the written answer so every run looks the same.

What this report does not claim

These 18 questions are from our practice files. This is not a public benchmark.